

IFN647 Week 6 Workshop

Language Models

In this workshop, you are going to design two language models by using inverted indexing structure. For a given query Q , a language model is going to find the relevant documents in a given folder, e.g., a folder "Test_Doc". Assume the documents in the folder are indexed by the python function `index_docs()`, i.e., `index_docs("Test_Doc", stop_words)` which returns a dictionary with the following data structure:

```
{term: {docID1: frequency1, DocID2: frequency2, ...}, ...}
```

Task 1. Design function `index_docs()` to construct an inverted index, a dictionary

```
{term:{docID1:freq1, DocID2:freq2, ...}, ...}.
```

For each document in the "Test_Doc" folder, it firstly finds the "docid" in tag <newsitem>. For each index term (excluding number, punctuations, and tags <p> and </p>) in <text>, it aslo inserts the term into the index or accumulates its frequency.

Task 2: Let $Q = \{q_1:1, q_2:1, \dots, q_n:1\}$ be a dictionary, please define a function `likelihood_IR(I, Q)` to estimate $P(Q|D)$, i.e., it returns the score of document D for the given query Q by using

$$P(Q|D) = \prod_{i=1}^n P(q_i|D)$$

$$P(q_i|D) = \frac{f_{q_i,D}}{|D|}$$

where $f_{q_i,D}$ is the number of times word q_i occurs in document D , and $|D|$ is the number of words in D .

Task 3: Assume you have obtained another folder, e.g., "Collection_C". Extend function `likelihood_IR(I, Q)` to a Jelinek-Mercer smoothing function `likelihood_JM(I_C, I_D, Q, lambda)` by using the following equation:

$$P(Q|D) = \prod_{i=1}^n \left((1 - \lambda) \frac{f_{q_i, D}}{|D|} + \lambda \frac{c_{q_i}}{|C|} \right)$$

where c_{q_i} is the number of times query term q_i occurs in I_C , and $|C|$ is the total number of term occurrences in I_C . See more details in page 12 of week 5 lecture notes.

The function estimates a score for documents in I_D .

You can assume that both the query and the document have the same text preprocessing applied. So, each word in D is a stem, and $|D|$ and $|C|$ simply count the number of terms (stems) in D or C (including repeated occurrences).

Task 4: Test the defined functions using the provided set of test documents (in folder "Test_docs") and a collection of documents (in folder "collection_C"). You need to use the 'common-english-words.txt' when indexing both sets of documents into `Index_Test` and `Index_C`, respectively. You can then test functions `likelihood_IR(Index_Test, Query)` and `likelihood_JM(Index_C, Index_Test, Query, lamda)` (you can assume `lamda = 0.4`, and `Query = {'formula':1, 'one':1}, {'share':1}` or `{'leaderboard':1, 'british':1}`).